Geetha Charan

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Education

Indian Institute of Science, Bangalore — **Aug 2022 - Jun 2024**
M.Tech in Computer Science and Automation

Jawaharlal Nehru Technological University, Anantapur — **Jun 2018 - Jun 2022**
B.Tech in Computer Science and Engineering

Work Experience

Research Associate | *Indian Institute of Science* — **Feb 2026 – Present**

- Leading AI research at Game Theory Lab under Professor Y. Narahari, bridging NLP, Deep Learning, eXplainable AI, and cooperative game theory.
- **Agricultural LLMs:** Fine-tuning foundation models, hybrid Knowledge Graphs (KG) and RAG architectures for domain-specific agricultural decision support systems.
- **Price Forecasting:** Foundational models and custom deep learning architectures for highly volatile crop price prediction; the work also explores LLMs as zero-shot time series predictors.
- **Explainable AI:** Developing novel cooperative game theory frameworks for model interpretability, investigating solution concepts beyond Shapley values - Core, Nucleolus, Gately Point, and hybrid approaches.
- **Carbon Farming:** Applying cooperative game theory and optimization to carbon pratice selection, coalition formation and market design for voluntary carbon markets.

Founding AI Researcher | *Finrep AI* — **Jun 2024 – Jan 2026**

- Led AI research team responsible for end-to-end ML infrastructure including indexing and chunking strategies, retrieval systems, LLM architecture design, verification layers, evaluation frameworks, and custom model development for financial intelligence.
- **Multi-Agent Systems:** Built a multi-agent system for financial disclosure generation with specialized agents for content drafting and regulatory compliance; developed an agentic retrieval-based answering system with tool calling and ReAct loop capabilities for financial queries.
- **Deep Learning for Compliance:** Designed dual BERT encoder architecture with custom cross-encoder for financial document compliance scoring, incorporating pairwise interaction modeling and masked max pooling; developed BERT-based classification architecture achieving over 95% accuracy on benchmark datasets.
- **LLM Engineering:** Fine-tuned and benchmarked LLMs for financial document processing; built context-aware report generation systems with filterations using classic ML to reduce context rot.
- **Research & Innovation:** Directed multiple financial NLP proofs-of-concept including PR Q&A Generation, LLM-Knowledge Graph integration, text-to-SQL, Invoice Processing System, and Prompt Induction.

Machine Learning Intern | *Accenture* — **Jun 2023 – Jul 2023**

- Implemented advanced classification models like XGBoost and LightGBM for methane emission detection, optimizing their performance through dataset scaling and efficient feature engineering.
- Addressed data scarcity by employing oversampling and synthetic data generation techniques to enhance the diversity and volume of training data, ultimately improving the models ability to generalize across different scenarios.

Publications

[1] **Geetha Charan** et al. *A Cooperative Game Theoretic Approach to Sustainable Carbon Farming.* ***(under review)***

[2] Vedant B, **Geetha Charan** et al. *Agri-SAGE: Simulation-Grounded Multi-Agent LLM for Context-Aware Agricultural Advisory Generation.* ***(under review)***

[3] Priyanka V, **Geetha Charan** et al. *Advancing Crop Planning with Soil and Irrigation-Aware Optimization Across Spatial Granularities.* ***(under review)***

[4] Priyanka V, **Geetha Charan** et al. *Carbon Farming: An Expository, Inter-Disciplinary Survey.* Journal of IISc, Special Issue on Digital Agriculture, 2026, Springer. [online]

Master's Thesis

Agent Based LLMs and Finetuning LLMs | **Game Theory Lab IISc** — ***Prof. Narahari***

- Developed an LLM-powered agentic interface for an end-to-end ML platform, enabling natural language-driven function calls for data analysis and model building via a fine-tuned function-calling model.
- Designed and deployed a domain-adapted QA system by fine-tuning LLaMa 3.1 8B, TinyLlama 1.1B on a curated research corpus and with FAISS-powered RAG pipeline.

ML Platform for Agricultural Data Analysis | **Game Theory Lab IISc** — ***Prof. Narahari***

- Designed and developed an end-to-end ML platform to streamline problem analysis in the field of Agriculture.
- The pipeline has various tools to help integrate, clean, imputate, and visualize the data. On top of that, it provides a diverse model selection that deals with tabular and time series datasets.

Relevant Coursework

- **Foundational Courses** : Natural Language Processing, Machine Learning, Game Theory, Computer Architecture.
- **Advanced Courses** : Advanced Deep Representation Learning, Advanced Image Processing.
- **Mathematical Courses** : Applied Linear Algebra and Optimization, Probability and Statistics.

Lectures / Presentations

- Guest Lecture, E1 254: Game Theory 2026, IISc
  taught Cooperative Game Theory solution concepts covering shapley values, SHAP, nucleolus and more.

Achievements / Responsibilities

- **All India Rank 15** in GATE Computer Science Exam 2022.
- **Teaching Assistant** E1 214: Game Theory and Mechanism Design; UE 101: Algorithms and Programming.